

JUSTIN DORJEE CORKE

(xxx) xxx-xxxx jdc2254@columbia.edu

EDUCATION

Columbia University

Master of Arts in Statistics, Theory & Methods Track

New York, NY

Spring 2026

- Admitted to the Theory & Methods honors track; passed Statistics PhD preliminary exam to qualify.
- **Relevant Coursework:** Analysis & Probability I (PhD); Applied Statistics I (PhD); Machine Learning, High-Dimensional Probability, Log-concave Probability

Middlebury College

Bachelor of Arts, Major in Mathematics, Minor in Computer Science

Middlebury, VT

February 2025

- awarded Summa Cum Laude

RESEARCH

MA Mentored Research: α -Potential Games for Mixed-Autonomy Traffic

New York, NY

Advisor: Professor Sharon Di

Spring 2026

- Extended α -potential games for coordinating autonomous vehicles (CAVs) from homogeneous fleets to mixed traffic, with human-driven vehicles (HDVs) modeled as a PDE density field.
- Derived a bound, under Lipschitz and PDE-stability assumptions, guaranteeing that minimizing a single potential function yields an approximate Nash equilibrium of the coupled CAV–HDV game.
- Built a Python simulation that couples neural-network CAV policies with a Lax–Friedrichs LWR solver via damped fixed-point iteration to empirically validate the bound.

Dynamic φ -Divergence Selection for Safe Exploitation in Leduc Hold'em

New York, NY

Co-authored with Shayan Ravari '27

Spring 2026

- Studied how the choice of divergence regularizer trades off exploiting weak opponents against staying unexploitable in imperfect-information Leduc Hold'em.
- Swept 6 φ -divergences across 500 regularization strengths to trace Pareto frontiers; derived an exploitability bound whose constants matched the empirical ranking.
- Framed adaptive (φ, λ) selection as a 30-arm Thompson Sampling bandit (20 seeds $\times$ 1,000 rounds); showed the optimal divergence shifted from Jensen-Shannon (19/20 seeds) to Forward KL and Chi-Squared as risk aversion rose.

Research Assistant, Network Science

Middlebury College, VT

Advisor: Professor Philip Chodrow

Summer 2024–Spring 2025

- Derived theoretical bounds on degree assortativity under graph union operations, validated with Monte Carlo simulations (Python, NetworkX) across Erdős–Rényi, Barabási–Albert and configuration-model graphs.

PROJECTS

Century: Golem Edition — Optimal-Play Solver

[GitHub]

- Built a Python engine for the board game Century as a simulation platform (with multiplayer web app) for strategy research.
- Tested which strategic factors drive winning by tuning an interpretable 12-feature bot with CMA-ES over 49,300 simulated games. It won 82% of games against a hand-tuned baseline and found that opponent modeling had no significant effect.
- Bounded the fewest turns to win in solo play; game heuristics and an integer program gave lower bounds, while beam search found near-optimal plans. On reduced goals where the optimum was provable via depth-first branch and bound (DFBB), the plans were within 0.1 turns of it.

SKILLS

Programming: Python (pandas, NumPy, SciPy, statsmodels, PyTorch, NetworkX), R, MATLAB

Methods: Statistical inference, regression, Bayesian, Monte Carlo simulation, machine learning (neural networks, bandits), optimization (convex, integer programming), game theory

Tools: Git, Jupyter, LaTeX, Claude Code